

Review Article

Coronary artery calcium in 2023: Guidelines for LDL-C goals, non-statin therapies, and aspirin use

Garshasb P. Soroosh^{a,*}, Erfan Tasdighi^a, Rishav Adhikari^b, Michael J. Blaha^a^a Johns Hopkins Ciccarone Center for the Prevention of Cardiovascular Disease, Johns Hopkins University School of Medicine, Baltimore, MD, USA^b Department of Medicine, Stanford University School of Medicine, Stanford, CA, USA

ARTICLE INFO

Keywords:

ASCVD

CAC

Lipid-lowering therapy.

Aspirin.

Personalized medicine.

ABSTRACT

Personalizing risk assessment and treatment decisions for the primary prevention of atherosclerotic cardiovascular disease (ASCVD) rely on pooled cohort equations and increasingly coronary artery calcium (CAC) score. A growing body of evidence supports that elevated CAC scores correspond to progressively elevated ASCVD risk, and that scores of ≥ 100 , ≥ 300 , and ≥ 1000 denote risk that is equivalent to certain secondary prevention populations. This has led consensus guidelines to incorporate CAC score thresholds for guiding escalation of preventive therapy for lowering low-density lipoprotein cholesterol goals, initiation of non-statin lipid lowering medications, and use of low-dose daily aspirin. As data on CAC continues to grow, more decision pathways will incorporate CAC score cutoffs to guide management of blood pressure and cardiometabolic medications. CAC score is also being used to enrich clinical trial study populations for elevated ASCVD risk, and to screen for subclinical coronary atherosclerosis in patients who received chest imaging for other diagnostic purposes.

Introduction

Since their introduction in 2013, the pooled cohort equations (PCE) have served as the primary tool to estimate 10-year atherosclerotic cardiovascular disease (CVD; ASCVD) risk and subsequently guide risk-reduction via lifestyle modifications and medical therapy.¹ The PCE include demographic variables of age, sex, race, and individual variables of smoking status, systolic blood pressure, hypertension treatment status, diabetes status, total and high-density lipoprotein cholesterol (HDL-C) levels to arrive at a risk estimate. In adults who are aged 40–75 years old, have a low-density lipoprotein cholesterol (LDL-C) level of 70–190 mg/dL, and do not have diabetes mellitus (DM), the 10-year ASCVD risk is subsequently categorized as Low Risk (<5%), Borderline Risk (5 to <7.5%), Intermediate Risk (≥ 7.5 to 20%), and High Risk ($\geq 20\%$).² Clinical guidelines for the primary prevention of ASCVD offer varied

recommendations for individualized risk discussion, assessment for presence of risk enhancers, and statin therapy for each PCE-defined risk group.

The ability to directly measure atherosclerotic coronary artery calcium (CAC) deposition using non-contrast computed tomography (CT) was first described by Agatston and Janowitz in 1990.³ At that time, given a lack of data demonstrating the clinical implications of CAC scoring for estimating ASCVD burden, this technique had little immediately actionable value to clinicians. Modern CAC scoring is conducted using non-contrast cardiac-gated multi-detector CT images of the chest, providing a score reported as an absolute value in Agatston units (AU) and as a percentile specific to age, sex, and race/ethnicity, as well as qualitative information on the relative distribution of calcium.^{4,5} CAC testing provides a rapid, sensitive, non-invasive, and low radiation method of visualizing subclinical ASCVD burden.⁶ Decades of clinical

Abbreviations: ACC, American College of Cardiology; AHA, American Heart Association; ASCVD, atherosclerotic cardiovascular disease; AU, Agatston units; BP, blood pressure; CAC, coronary artery calcium; CAC-DRS, Calcium Data and Reporting System; CHD, coronary heart disease; CT, computed tomography; CVD, cardiovascular disease; DM, diabetes mellitus; FH, familial hypercholesteremia; GIP, glucose-dependent insulinotropic polypeptide; GLP1a, glucagon-like peptide-1 receptor agonists; HDL-C, high-density lipoprotein cholesterol; hsCRP, high-sensitivity C-reactive protein; LDL-C, low-density lipoprotein cholesterol; Lp(a), lipoprotein (a); MACE, major adverse cardiovascular events; MESA, Multi-Ethnic Study of Atherosclerosis; MI, myocardial infarction; NLA, National Lipid Association; PCE, pooled cohort equations; PCSK9, proprotein convertase subtilisin kexin type 9; SCCT, Society of Cardiovascular CT; SGLT2i, sodium-glucose cotransporter 2 inhibitors.

* Corresponding author at: Johns Hopkins University School of Medicine, Ciccarone Center for the Prevention of Cardiovascular Disease, 600 N Wolfe St, Blalock 524D1, Baltimore, MD 21287, USA.

E-mail address: gsoroos1@jhmi.edu (G.P. Soroosh).

<https://doi.org/10.1016/j.pcad.2024.05.004>

Available online 15 May 2024

0033-0620/© 2024 Elsevier Inc. All rights reserved, including those for text and data mining, AI training, and similar technologies.

research has elucidated that elevated CAC burden is strongly associated with elevated risk for development of clinical ASCVD, including CVD, coronary heart disease (CHD), myocardial infarction (MI), stroke, cardiac and all-cause mortality.^{7–10} This risk persists across a range of prediction time periods and populations.¹¹ Additionally, CAC score of zero reliably predicts low risk of ASCVD.^{12,13} Quantification of sub-clinical coronary artery atherosclerosis via CAC is currently the most personalized ASCVD risk assessment tool in asymptomatic individuals, especially when compared to the traditional PCE.⁷ One novel risk assessment tool under investigation is the polygenic risk score, however, it does not significantly improve risk reclassification as CAC score does, and inferior risk discrimination compared to CAC score.¹⁴

Until recently, the evidence-based clinical use of CAC was to help determine whether patients at borderline or intermediate ASCVD risk based on PCE should be initiated on a statin. The 2019 ACC/AHA guidelines on CVD primary prevention only highlight this very simple use of CAC.² While guiding statin use remains a fundamental role of CAC, newer data has expanded its role beyond just statin therapy decisions to help guide personalized initiation of other lipid lowering therapies, antiplatelet therapy, and blood pressure management. It is now commonly understood that lowering LDL-C leads to proportionally lower ASCVD risk, and guidelines reflect lower LDL-C goals in those who are higher risk. The SPRINT trial demonstrated that more intensive blood pressure (BP) improves CVD outcomes, favoring more aggressive management in those with elevated ASCVD risk.¹⁵ Various trials suggest that daily aspirin for primary prevention based on traditional risk factors has borderline benefit and associated harm from bleeding, making CAC score a valuable tool in helping determine which patients will obtain net benefit from antiplatelet therapy.^{16,17}

The latest major primary prevention consensus guidelines outlined by the American Heart Association (AHA), American College of Cardiology (ACC), National Lipid Association (NLA), and others incorporate CAC scoring to reflect these expanded uses. For example, in contrast to the 2019 AHA/ACC guidelines, the 2022 ACC Expert Consensus Decision Pathway on the Role of Nonstatin Therapies for LDL-Cholesterol Lowering provides guidelines beyond just statins to other lipid lowering therapies including ezetimibe and proprotein convertase subtilisin kexin type 9 (PCSK9) inhibitors.¹⁸ These guidelines include for the first time explicit LDL-C treatment goals for primary prevention stratified by CAC score. The 2020 NLA Scientific Statement on CAC Scoring to Guide Preventive Strategies for ASCVD Risk outlines an even broader role for CAC across a range of adult age groups, within various high risk special populations, including DM and familial hypercholesterolemia (FH), and for BP management and aspirin therapy.¹⁹

Using CAC score to guide LDL-C goals and non-statin therapy

Statins remain the mainstay of LDL-C lowering therapy for ASCVD risk reduction. However, high intensity statins have a maximum LDL-C lowering effect of 50% on average, meaning additional lowering when indicated needs to be achieved with non-statin add-on therapy. Currently ezetimibe and PCSK9 inhibitors (monoclonal antibodies and siRNA) are the second and third therapies respectively recommended by the guidelines as stepwise additions to further decrease LDL-C. When taken in combination with statin therapy, ezetimibe decreases LDL-C by a mean of 20–25%, PCSK9 monoclonal antibodies (alirocumab and evolocumab) decrease LDL-C by a mean of 45–64% depending on dose and frequency, and PCSK siRNA (inclisiran) decreases LDL-C by a mean of 50%.¹⁸ Other LDL-C lowering agents include bile acid sequestrants and bempedoic acid, which has been shown to reduce LDL-C and major adverse cardiovascular events (MACE) in statin intolerant patients randomized in the CLEAR Outcomes trial.²⁰ It has become clear that evidence guided initiation of statins and non-statin therapies ultimately requires appropriate risk assessment through ASCVD and CAC scoring.

For adults who are Borderline Risk and Intermediate Risk (10-year ASCVD risk of 5% to <20%), consensus guidelines recommend assessing

for the presence of risk enhancing factors such as family history of premature ASCVD, persistent elevation of LDL-C ≥ 160 mg/dL, chronic kidney disease, metabolic syndrome, conditions specific to women (preeclampsia, premature menopause), chronic inflammatory diseases, South Asian ethnicity, persistently elevated non-fasting triglycerides ≥ 175 mg/dL, and in select individuals if measured high-sensitivity C-reactive protein (hsCRP) ≥ 2 mg/L, lipoprotein (a) [Lp(a)] ≥ 50 mg/dL or 125 nmol/L, apo B > 130 mg/dL, and ankle-brachial index < 0.9 .^{2,19} The presence or absence of these factors can help guide risk discussion and ultimately recommendation for initiating statin therapy. However, the amount of risk increase with each of these factors is unclear, and in the prior guidelines there still remained uncertainty about recommendation of statin intensity, LDL-C goal, and addition of non-statin therapy.¹⁹ The 2013 ACC/AHA cholesterol guideline did not recommend specific LDL-C goals in primary prevention and listed this as a “critical question” to be addressed in future studies.²¹ The 2022 ACC Expert Consensus Decision Pathway updates the prior guideline and provides discrete recommendations for discrete CAC score ranges based on updated data which are summarized below.¹⁸

Regardless of the presence of risk-enhancing factors, CAC score of zero confers reduced risk.¹² In those with CAC score of 0 statin initiation can be deferred until CAC score is remeasured in 3–5 years, unless if a high-risk condition is present (DM, LDL-C ≥ 190 mg/dL, family history of premature CHD, or cigarette smoking) in which case statin therapy is recommended.¹⁸ For CAC score of 1 to 99 AU and < 75 th percentile for that individual's age/sex/race, initiation of moderate intensity statin is recommended to achieve an LDL-C goal of < 100 mg/dL, with increase to high intensity statin if the goal is not achieved.¹⁸ For CAC score of ≥ 100 AU or ≥ 75 th percentile for age/sex/race, moderate to high intensity statin is recommended to achieve an LDL-C goal of < 70 mg/dL, with consideration to add ezetimibe if goal is not achieved.¹⁸ For CAC score of ≥ 1000 AU the recommendation is strengthened further to high intensity statin, followed by stepwise addition of ezetimibe and PCSK9 inhibitor therapy if LDL-C goal of < 70 mg/dL is not met.¹⁸ This distinction was made to appropriately escalate treatment intensity in individuals with very high CAC (≥ 1000 AU) as their risk for ASCVD events and mortality is commensurate to that of stable secondary prevention populations.¹⁰ This guideline provides clear pathways by which CAC score can personalize statin treatment, non-statin LDL-C lowering therapies, and LDL-C treatment goals to a greater extent than PCE alone.

These recommendations and in particular the stepwise intensification of cholesterol management at CAC ≥ 100 and CAC ≥ 1000 are largely based on the finding that ASCVD mortality and MACE in primary prevention individuals with substantive CAC can approach and equal that of stable secondary prevention, suggesting benefit from similar treatment goals (Fig. 1). An analysis of the Multi-Ethnic Study of Atherosclerosis (MESA) cohort found that CAC score of 0 corresponded to a 10-year ASCVD risk of less than 5%, while a CAC score of ≥ 100 increases risk to at least 7.5%.²² In an analysis comparing annualized ASCVD mortality in the CAC Consortium cohort with the FOURIER trial, CAC score of ≥ 300 in a primary prevention population was of similar risk as stable treated secondary prevention subpopulation with a single previous MI event.⁸ There is a dose-dependent increase in ASCVD mortality risk with increase in CAC, with a CAC score of 775–900 for non-diabetic primary prevention population and CAC score of 300–375 for primary prevention patients with diabetes having risk equivalent to secondary prevention patients in FOURIER.⁸ Compared to those with no CAC, a CAC score of ≥ 1000 increased the hazard ratio for CVD 6.8-fold and all-cause mortality 2.9-fold, a significant increase from those with CAC scores of 400–999, in a study of the CAC Consortium.¹² This finding is evident in the MESA cohort also, with CAC score ≥ 1000 conferring higher risk for cardiovascular disease events, non-cardiovascular disease outcomes, and mortality than lower CAC, and corresponding to an annualized MACE rate similar to the secondary prevention population of FOURIER, and higher than lower-risk FOURIER subpopulations.¹⁰ Given continued evidence that lower LDL-C lowers CVD risk, these data

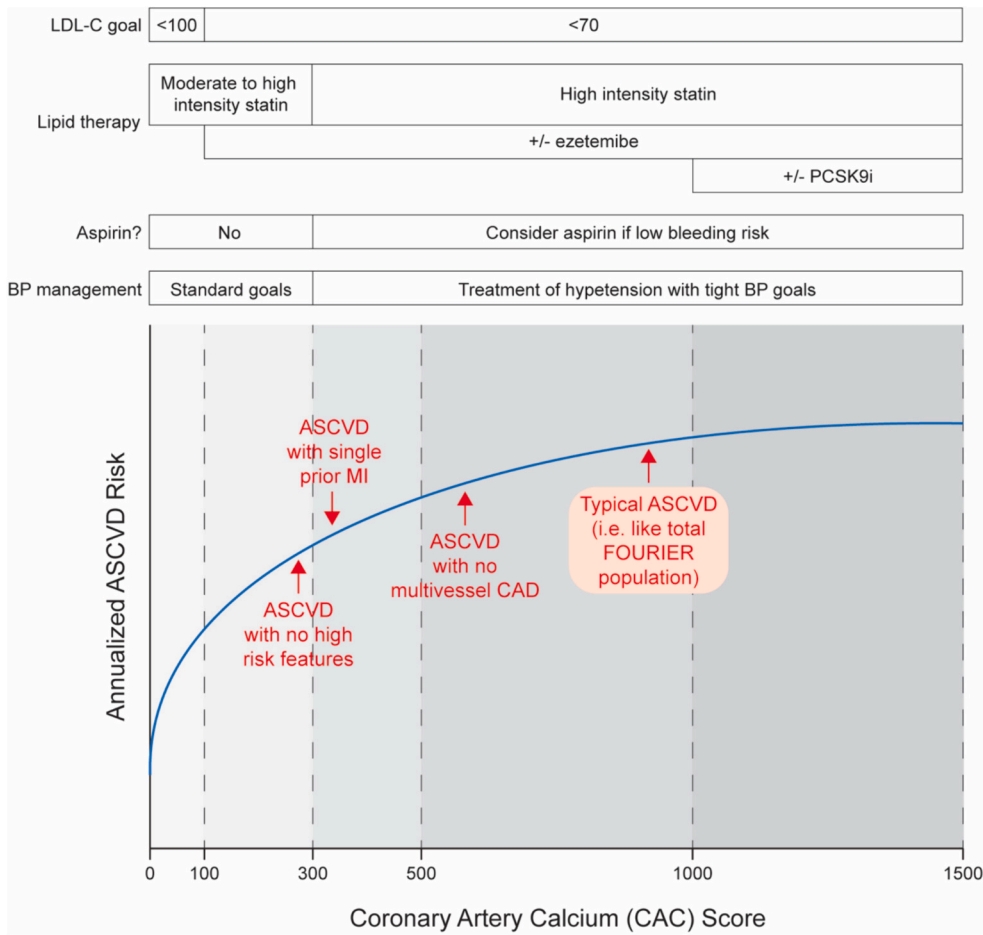


Fig. 1. Representative illustration of CAC score correlation with annualized ASCVD risk. Red arrows denote corresponding risk of notable secondary prevention populations with known ASCVD. LDL-C goal, lipid therapy, low-dose daily aspirin, and blood pressure (BP) management consensus guidelines are summarized above. Column dividers in the table correspond to CAC score values noted on graph axis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

provide impetus for the updated guidelines to lower LDL-C treatment targets with increasing CAC scores.²³

Using CAC score to guide aspirin therapy

The use of aspirin for primary prevention of ASCVD has recently undergone scrutiny.²⁴ Various guidelines have provided conflicting advice over the years, with some recommending universal use in certain age groups, while others advocating for a more selective approach based on individual risk factors.^{16,24} While aspirin reduces the risk of cardiovascular events through inhibition of platelet aggregation, it also increases the risk of bleeding, especially gastrointestinal bleeding.^{16,17} Randomized control trials and meta-analyses have provided conflicting evidence about the net benefit-risk ratio of aspirin use in primary prevention, leading to changes in clinical guidelines over time.^{2,17} Considering these controversies there is a strong rationale for a more personalized approach to aspirin therapy in primary prevention.

One strategy to optimize the use of aspirin in primary prevention is to better identify individuals at high risk of ASCVD who are not at increased risk of bleeding.² However, traditional ASCVD risk scoring also has a positive correlation with bleeding risk, meaning individuals with higher ASCVD risk as measured by PCE who would benefit the most from aspirin therapy are also often higher bleeding risk.²⁵ CAC scoring is one tool that can further personalize ASCVD risk assessment to guide aspirin therapy. While PCE alone is unable to differentiate individuals in whom aspirin is of net benefit, quantifying the burden of subclinical

atherosclerosis via CAC score can allow for tailored aspirin use in those with substantial CAC who are not high bleeding risk, and avoidance in those with minimal CAC. Importantly, high CAC scores are associated with increased ASCVD risk but are not directly linked to bleeding risk, making CAC particularly valuable for decision-making regarding aspirin therapy.²⁶

Based on the available evidence, the NLA recommends considering aspirin therapy for primary prevention in individuals with a CAC score >100, in the absence of high bleeding risk.¹⁹ This approach allows for a more personalized risk-benefit assessment, targeting aspirin therapy to only those who would benefit the most and are otherwise low bleeding risk.¹⁹ This strategy allows for individualization of aspirin therapy to each patient.

Several studies support this approach. Miedema et al. found that participants in the MESA cohort with a CAC score of 100 or higher taking daily low-dose aspirin were estimated to be 2 to 4 times more likely to prevent a heart attack than to experience a major bleed.²⁷ Participants with CAC score of 0 taking daily aspirin were estimated to be 2 to 4 times more likely to suffer a major bleed than to prevent a heart attack. Of note, the results of this study were independent of whether individuals qualified for aspirin use by the American Heart Association guidelines at the time. Moreover, although the authors consider differences in bleeding rates based on factors such as sex and age, the estimated increased risk of bleeding associated with aspirin use was assumed to be the same across all subgroups. These initial results highlighted the potential capacity of CAC score to guide aspirin therapy for primary

prevention.

Following up on the Miedema et al. study, another study of the MESA cohort by Cainzos-Achirica et al. reported that CAC score of 100 or higher, and in particular 400 or higher, demonstrates a net benefit of aspirin use in asymptomatic individuals below 70 years old who are not high bleeding risk.²⁸ While PCE alone is unable to differentiate individuals in whom aspirin is of net benefit, quantifying the burden of subclinical atherosclerosis via CAC score appears to allow for tailored aspirin use in those with substantial CAC who are not high bleeding risk, and avoidance in those with minimal CAC. For all estimated ASCVD risk groups, CAC ≥ 100 identified individuals in whom the CVD benefit of aspirin is greater than the increased bleeding risk.²⁸ This was particularly true in younger individuals, attributable to their lower bleeding risk. CAC ≥ 400 in particular identified those in whom the net benefit of aspirin was most pronounced, and CAC of zero identified those in whom increased bleeding risk outweighed CVD benefit, regardless of estimated ASCVD risk using PCE.²⁸ In primary prevention patients CAC score can guide personalized initiation of low-dose aspirin therapy, and it is our view that future ACC/AHA prevention guidelines should follow the NLA and consider incorporating CAC-based recommendations for aspirin therapy.

Conclusions

CAC scoring is more than a tie breaker for initiating statin therapy. The CAC score quantifies the degree of subclinical atherosclerosis present, providing the most sensitive and personalized ASCVD risk estimation currently available for the asymptomatic primary prevention population. A personalized medicine approach to ASCVD risk assessment and treatment starts with risk estimation with PCE, and subsequently with CAC scoring in many. In the updated 2023 ACC/AHA Multimodality Appropriate Use Criteria for the Detection and Risk Assessment of Chronic Coronary Disease, CAC score is now the only testing modality indicated for all asymptomatic patients without known ASCVD.²⁹ As data on harnessing CAC to personalize preventive care continues to grow, so will its incorporation into further consensus guidelines and clinical decision pathways.

Future directions

Personalizing allocation of novel therapies and treatment goals

Similar to its role in guiding the use of LDL-C lowering therapy and low-dose aspirin, CAC scoring has the potential to personalize allocation of newer generations of cardiometabolic therapies including sodium-glucose cotransporter 2 inhibitors (SGLT2i), glucagon-like peptide-1 receptor agonists (GLP1a), and newer glucose-dependent insulinotropic polypeptide (GIP) agonist therapies. Currently these cardioprotective medications are only indicated in those who have specific comorbidities such as heart failure, chronic kidney disease, diabetes mellitus, or obesity. However, these medications are costly and will not be able to be provided to all patients with these indications, particularly those with obesity. In the future significant subclinical atherosclerosis as demonstrated by CAC could serve as an additional criterion for initiating therapy and ensuring insurance coverage and reimbursement.³⁰ Similarly, consensus guidelines are incorporating personalized blood pressure targets based on CAC score, with more intensive goals being recommended for those with higher atherosclerotic burden similar to guidelines in the secondary prevention population.¹⁹ For example, a CAC score of ~ 250 provides risk equivalent to those in the SPRINT trial, pointing to the potential benefit of tight BP goals in this high risk population.³¹

Additionally, CAC burden has utility to re-stratify risk among patient populations that are currently universally classified as elevated risk.³² For example, while DM increases ASCVD risk on a population level and guidelines recommend statin use for primary prevention, data from the Walter Reed Cohort Study suggests that 52% of adults with DM have

CAC=0, and this subgroup may have relatively lower CVD risk and consequently may not derive as much benefit from intensive lipid lowering approaches.⁹ Such studies exploring the de-risking potential of CAC=0 in subpopulations such as mild DM, obesity, and severe hypercholesterolemia may help direct treatment strategies to those who can most benefit while avoiding unnecessary treatment in those who can be re-classified as low risk.

Enriching populations in clinical trials

Given the specificity of CAC in identifying high-risk and low-risk individuals, the CAC score is increasingly being suggested for use in randomized control trials to identify those patients most likely to benefit from interventions.¹¹ Because average event rates in the primary prevention population are much lower than in the secondary prevention population, even among cohorts with high estimated risk based on PCE, randomized controlled trials for primary CVD prevention tend to be particularly challenging. To be sufficiently powered, these trials require very large study populations and long follow up periods that lengthen the pace of data collection, add additional cost, and often prohibit initiation of new studies. CAC has the potential to mitigate these barriers by allowing for study designs that enrich for participants with high CAC scores and therefore more likely to experience a cardiovascular event and exclude participants with CAC = 0 who are unlikely to have events. A simulation of this enrichment strategy using the MESA cohort estimated that employing high CAC as a study entry criterion could significantly save study costs by lowering the necessary cohort size of a typical primary ASCVD prevention trial up to 58%.³³ While no large primary prevention cardiovascular outcomes trial to date has relied purely on a CAC based enrichment strategy, several prominent recent and ongoing trials have incorporated CAC alongside other risk enhancing factors to identify elevated risk individuals for inclusion.³⁴ One example is the ongoing VESALIUS-CV trial (Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke). Such a strategy may have additional impact if the Lp(a) lowering drug pelacarsen is successful at lowering ASCVD events in secondary prevention in the ongoing Lp(a) HORIZON trial.

Opportunistic screening for CAC

Another area of interest is assessing CAC and calculating CAC scores from non-gated CT chest imaging obtained for other diagnostic purposes. In these cases, CAC is reported as an incidental finding on the imaging study. Timely identification of incidental CAC can trigger initiation of preventive therapies in those who otherwise would have gone without risk reducing treatment. Such a screening strategy has the potential to profoundly expand the population of patients evaluated for subclinical coronary atherosclerosis; approximately 80 million non-gated chest CT scans are performed annually in the U.S., compared to just 2 million cardiac CT scans.¹¹ The reporting of CAC burden from non-cardiac CT scans is endorsed by the Society of Cardiovascular CT (SCCT), which recommends using the Coronary Artery Calcium Data and Reporting System (CAC-DRS), a standardized scoring system validated in ECG-gated CAC scans that can also be applied to quantify CAC in non-gated CT scans.^{35,36} In addition to interpreting CAC seen on new CT imaging, prior CT images available in the electronic medical record can be retroactively reviewed for presence and degree of CAC as a broad screening intervention. In these cases of opportunistic screening for CAC, artificial intelligence algorithms can provide automated and cost-effective screening, increasing the number of scans that can be evaluated and patients served by this approach.³⁷ This strategy was recently demonstrated in the NOTIFY-1 trial, in which a deep learning algorithm was used to screen for incidental CAC in prior chest CT images and identified 173 patients with evidence of CAC as confirmed by a radiologist.³⁷ Half of these patients were randomly assigned to have their primary care physician receive a notification of this finding and the ACC/AHA guideline recommendations regarding statin use in patients with CAC. Those in the intervention arm were over seven times more

likely than the non-notified group to have initiated a statin within the next six months (51% vs 7%). A subsequent study, NOTIFY-2 (NCT05394181), is currently underway to also test this strategy in a larger patient cohort.

CRedit authorship contribution statement

Garshasb P. Soroosh: Supervision. **Erfan Tasdighi:** Supervision. **Rishav Adhikari:** Supervision. **Michael J. Blaha:** Supervision.

Declaration of competing interest

The authors have no conflicts of interest to disclose. E. Tasdighi has received support from National Institutes of Health (Grant T32 HL007227).

References

- Goff Jr DC, Lloyd-Jones DM, Bennett G, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association task force on practice guidelines. *J Am Coll Cardiol.* 2014;63(25 Pt B):2935–2959.
- Arnett DK, Blumenthal RS, Albert MA, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines. *Circulation.* 2019;140(11):e596–e646.
- Agatston AS, Janowitz WR, Hildner FJ, Zusmer NR, Viamonte Jr M, Detrano R. Quantification of coronary artery calcium using ultrafast computed tomography. *J Am Coll Cardiol.* 1990;15(4):827–832.
- Alluri K, Joshi PH, Henry TS, Blumenthal RS, Nasir K, Blaha MJ. Scoring of coronary artery calcium scans: history, assumptions, current limitations, and future directions. *Atherosclerosis.* 2015;239(1):109–117.
- Obisesan OH, Osei AD, Uddin SMI, Dzaye O, Blaha MJ. An update on coronary artery calcium interpretation at chest and cardiac CT. *Radiol Cardiothorac Imag.* 2021;3(1), e200484.
- Nasir K, Cainzos-Achirica M. Role of coronary artery calcium score in the primary prevention of cardiovascular disease. *Bmj.* 2021;373, n776.
- Miedema MD, Dardari ZA, Nasir K, et al. Association of Coronary Artery Calcium with Long-term, cause-specific mortality among Young adults. *JAMA Netw Open.* 2019;2(7), e197440.
- Dzaye O, Razavi AC, Michos ED, et al. Coronary artery calcium scores indicating secondary prevention level risk: findings from the CAC consortium and FOURIER trial. *Atherosclerosis.* 2022;347:70–76.
- Mitchell JD, Paisley R, Moon P, Novak E, Villines TC. Coronary artery calcium and long-term risk of death, myocardial infarction, and stroke: the Walter reed cohort study. *JACC Cardiovasc Imaging.* 2018;11(12):1799–1806.
- Peng AW, Dardari ZA, Blumenthal RS, et al. Very high coronary artery calcium (≥ 1000) and association with cardiovascular disease events, non-cardiovascular disease outcomes, and mortality: results from MESA. *Circulation.* 2021;143(16): 1571–1583.
- Whelton SP, Blaha MJ. Coronary artery calcium: from risk prediction to treatment allocation and clinical trials. *Heart.* 2023;109(22):1714–1721.
- Adelhoefer S, Uddin SMI, Osei AD, Obisesan OH, Blaha MJ, Dzaye O. Coronary artery calcium scoring: new insights into clinical interpretation-lessons from the CAC consortium. *Radiol Cardiothorac Imag.* 2020;2(6), e200281.
- Agha AM, Pacor J, Grandhi GR, et al. The prognostic value of CAC zero among individuals presenting with chest pain: a Meta-analysis. *JACC Cardiovasc Imaging.* 2022;15(10):1745–1757.
- Khan SS, Post WS, Guo X, et al. Coronary artery calcium score and polygenic risk score for the prediction of coronary heart disease events. *Jama.* 2023;329(20): 1768–1777.
- Wright Jr JT, Williamson JD, Whelton PK, et al. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med.* 2015;373(22):2103–2116.
- McNeil JJ, Wolfe R, Woods RL, et al. Effect of aspirin on cardiovascular events and bleeding in the healthy elderly. *N Engl J Med.* 2018;379(16):1509–1518.
- Gaziano JM, Brotons C, Coppolecchia R, et al. Use of aspirin to reduce risk of initial vascular events in patients at moderate risk of cardiovascular disease (ARRIVE): a randomised, double-blind, placebo-controlled trial. *Lancet.* 2018;392(10152): 1036–1046.
- Lloyd-Jones DM, Morris PB, Ballantyne CM, et al. 2022 ACC expert consensus decision pathway on the role of nonstatin therapies for LDL-cholesterol lowering in the Management of Atherosclerotic Cardiovascular Disease Risk: a report of the American College of Cardiology Solution set Oversight Committee. *J Am Coll Cardiol.* 2022;80(14):1366–1418.
- Orringer CE, Blaha MJ, Blankstein R, et al. The National Lipid Association scientific statement on coronary artery calcium scoring to guide preventive strategies for ASCVD risk reduction. *J Clin Lipidol.* 2021;15(1):33–60.
- Nissen SE, Lincoff AM, Brennan D, et al. Bempedoic acid and cardiovascular outcomes in statin-intolerant patients. *N Engl J Med.* 2023;388(15):1353–1364.
- Stone NJ, Robinson JG, Lichtenstein AH, et al. 2013 ACC/AHA guideline on the treatment of blood cholesterol to reduce atherosclerotic cardiovascular risk in adults: a report of the American College of Cardiology/American Heart Association task force on practice guidelines. *J Am Coll Cardiol.* 2014;63(25 Pt B):2889–2934.
- Budoff MJ, Young R, Burke G, et al. Ten-year association of coronary artery calcium with atherosclerotic cardiovascular disease (ASCVD) events: the multi-ethnic study of atherosclerosis (MESA). *Eur Heart J.* 2018;39(25):2401–2408.
- Wang N, Woodward M, Huffman MD, Rodgers A. Compounding benefits of cholesterol-lowering therapy for the reduction of major cardiovascular events: systematic review and Meta-analysis. *Circ Cardiovasc Qual Outcomes.* 2022;15(6), e008552.
- Bibbins-Domingo K. Aspirin use for the primary prevention of cardiovascular disease and colorectal cancer: U.S. preventive services task force recommendation statement. *Ann Intern Med.* 2016;164(12):836–845.
- Selak V, Kerr A, Poppe K, et al. Annual risk of major bleeding among persons without cardiovascular disease not receiving antiplatelet therapy. *Jama.* 2018;319 (24):2507–2520.
- Verghese D, Manubolu S, Budoff MJ. Contemporary use of coronary artery calcium for the allocation of aspirin in light of the 2022 USPSTF guideline recommendations. *Am J Prev Cardiol.* 2022;12, 100427.
- Miedema MD, Duprez DA, Misialek JR, et al. Use of coronary artery calcium testing to guide aspirin utilization for primary prevention: estimates from the multi-ethnic study of atherosclerosis. *Circ Cardiovasc Qual Outcomes.* 2014;7(3):453–460.
- Cainzos-Achirica M, Miedema MD, McEvoy JW, et al. Coronary artery calcium for personalized allocation of aspirin in primary prevention of cardiovascular disease in 2019: the MESA study (multi-ethnic study of atherosclerosis). *Circulation.* 2020;141 (19):1541–1553.
- Winchester DE, Maron DJ, Blankstein R, et al. ACC/AHA/ASE/ASNC/ASPC/HFSA/HRS/SCAI/SCCT/SCMR/STS 2023 multimodality appropriate use criteria for the detection and risk assessment of chronic coronary disease. *J Am Coll Cardiol.* 2023; 81(25):2445–2467.
- Cainzos-Achirica M, Patel KV, Quispe R, et al. Coronary artery calcium for the allocation of GLP-1RA for primary prevention of atherosclerotic cardiovascular disease. *JACC Cardiovasc Imaging.* 2021;14(7):1470–1472.
- Uddin SMI, Mirbolouk M, Kianoush S, et al. Role of coronary artery calcium for stratifying cardiovascular risk in adults with hypertension. *Hypertension.* 2019;73 (5):983–989.
- Blaha MJ, Cainzos-Achirica M, Greenland P, et al. Role of coronary artery calcium score of zero and other negative risk markers for cardiovascular disease: the multi-ethnic study of atherosclerosis (MESA). *Circulation.* 2016;133(9):849–858.
- Cainzos-Achirica M, Bittencourt MS, Osei AD, et al. Coronary artery calcium to improve the efficiency of randomized controlled trials in primary cardiovascular prevention. *JACC Cardiovasc Imaging.* 2021;14(5):1005–1016.
- Cainzos-Achirica M, Anugula D, Mszar R, et al. Rationale and pathways forward in the implementation of coronary artery calcium-based enrichment of randomized trials. *Am Heart J.* 2022;243:54–65.
- Dzaye O, Dudum R, Mirbolouk M, et al. Validation of the coronary artery calcium data and reporting system (CAC-DRS): dual importance of CAC score and CAC distribution from the coronary artery calcium (CAC) consortium. *J Cardiovasc Comput Tomogr.* 2020;14(1):12–17.
- Hecht HS, Cronin P, Blaha MJ, et al. 2016 SCCT/STR guidelines for coronary artery calcium scoring of noncontrast noncardiac chest CT scans: a report of the Society of Cardiovascular Computed Tomography and Society of thoracic radiology. *J Cardiovasc Comput Tomogr.* 2017;11(1):74–84.
- Sandhu AT, Rodriguez F, Ngo S, et al. Incidental coronary artery calcium: opportunistic screening of previous nongated chest computed tomography scans to improve statin rates (NOTIFY-1 project). *Circulation.* 2023;147(9):703–714.